

Day 12 — Custom Wazuh Rules

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Objective

Write 3 custom Wazuh detection rules addressing distinct scenarios: privileged file staging in temp directories, brute-force composite detection, and reverse shell indicators via auditd — standard detection engineering work performed during client engagements.

Deliverable 1-

Rule 1 — Privileged File Creation in /tmp

Objective: Detect a privileged user (root) creating a file in a world-writable temp directory — a common attacker staging pattern.

Final rule:

```
xml
<rule id="100200" level="10">
  <if_group>audit</if_group>
  <field name="audit.file.name" type="pcre2">^tmp\[^\]/]+$</field>
  <field name="audit.uid">0</field>
  <description>Privileged user (root) created a file in
/tmp</description>
  <mitre>
    <id>T1074</id>
  </mitre>
</rule>
```

```

fara@ubuntu:~$ cat > ~/day12-rule1.xml << 'EOF'
<group name="local,">
  <rule id="100200" level="10">
    <if_group>audit</if_group>
    <field name="audit.file.name" type="pcre2">^tmp\[^\]+$</field>
    <field name="audit.uid">0</field>
    <description>Privileged user (root) created a file in /tmp</description>
    <mitre>
      <id>T1074</id>
    </mitre>
  </rule>
</group>
EOF

docker cp ~/day12-rule1.xml single-node-wazuh.manager-1:/var/ossec/etc/rules/day12_rule1.xml

docker exec single-node-wazuh.manager-1 /var/ossec/bin/wazuh-control start
Successfully copied 368B (transferred 2.05kB) to single-node-wazuh.manager-1:/var/ossec/etc/
rules/day12_rule1.xml
2026/07/13 10:09:23 wazuh-modulesd:router: INFO: Loaded router module.
2026/07/13 10:09:23 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Starting Wazuh v4.9.0...
wazuh-apid already running...
Started wazuh-csyslogd...
Started wazuh-dbd...
2026/07/13 10:09:24 wazuh-integrator: INFO: Remote integrations not configured. Clean exit.
Started wazuh-integrator...
Started wazuh-agentlessd...
wazuh-authd already running...
wazuh-db already running...
wazuh-execd already running...
wazuh-analysisd already running...
wazuh-syscheckd already running...
wazuh-remoted already running...
wazuh-logcollector already running...
wazuh-monitor already running...
wazuh-modulesd already running...
Completed.

```

Key findings during development:

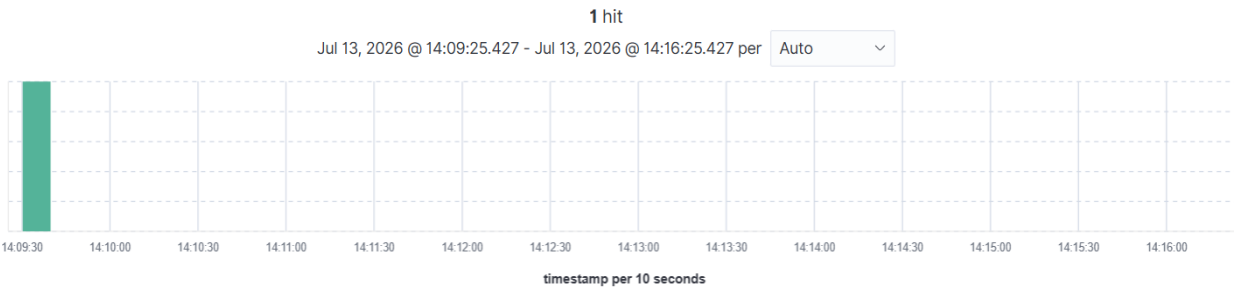
- Auditd was not configured to watch /tmp at all — the root cause of initial false results. Added a persistent watch rule: `sudo auditctl -w /tmp -p wa -k tmp_watch`.
- An initial loose regex (`.*\tmp\.*`) matched Wazuh's own internal path (`queue/vd_updater/tmp/contents/...`) as a false positive. Corrected to a fully-anchored pattern (`^tmp\[^\]+$`) requiring the field to consist *entirely* of tmp/ plus a filename.

Verification:

Confirmed via live agent data — `sudo touch /tmp/finaltest.txt` produced a clean, single, correctly-tagged alert with no false positives.

timestamp: Jul 13, 2026 @ 14:09:37.773 ×

+ Add filter



time

_source

Jul 13, 2026 @ 14:09:37.773

input.type: log agent.ip: 192.168.56.104 agent.name: agent-ubuntuserver agent.id: 004 manager.name: wazuh.manager

data.audit.syscall: 257 data.audit.gid: 0 data.audit.fsgid: 0 data.audit.session: 3 data.audit.pid: 11458

data.audit.suid: 0 data.audit.type: SYSCALL data.audit.directory.inode: 524292 data.audit.directory.mode: 041777

data.audit.directory.name: /tmp/ data.audit.uid: 0 data.audit.egid: 0 data.audit.exe: /usr/bin/touch

data.audit.file.inode: 579971 data.audit.file.mode: 0100644 data.audit.file.size: 1024 data.audit.file.type: file

@timestamp	Jul 13, 2026 @ 14:09:37.773
_index	wazuh-alerts-4.x-2026.07.13
agent.id	004
agent.ip	192.168.56.104
agent.name	agent-ubuntuserver
data.audit.arch	c000003e
data.audit.auid	1000
data.audit.command	touch
data.audit.cwd	/home/fara
data.audit.directory.inode	524292
data.audit.directory.mode	041777
data.audit.directory.name	/tmp/
data.audit.egid	0
data.audit.euid	0
data.audit.exe	/usr/bin/touch
data.audit.exit	3
data.audit.file.inode	579971
data.audit.file.mode	0100644

t full_log	type=SYSCALL msg=audit(1783937376.668:436227): arch=c000003e syscall=257 success=yes exit=3 a0=ffffff9c a1=7ffc825d18d6 a2=941 a3=1b6 items=2 ppid=11457 pid=11458 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts1 ses=3 comm="touch" exe="/usr/bin/touch" subj=unconfined key="tmp_watch"ARCH=x86_64 SYSCALL=openat AUID="fara" UID="root" GID="root" EUID="root" SUID="root" FSUID="root" EGID="root" SGID="root" FSGID="root" type=CWD msg=audit(1783937376.668:436227): cwd="/home/fara" type=PATH msg=audit(1783937376.668:436227): item=0 name="/tmp/" inode=524292 dev=fd:00 mode=041777 ouid=0 ogid=0 rdev=00:00 nametype=PARENT cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0OUID="root" OGID="root" type=PATH msg=audit(1783937376.668:436227): item=1 name="/tmp/finaltest.txt" inode=579971 dev=fd:00 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=CREATE cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0OUID="root" OGID="root" type=PROCTITLE msg=audit(1783937376.668:436227): proctitle=746F756368002F746D702F66696E616C746573742E747874	
t id	1783937377.1592266	
t input.type	log	
t location	/var/log/audit/audit.log	
t manager.name	wazuh.manager	
t rule.description	Privileged user (root) created a file in /tmp	
# rule.firedtimes	27	
t rule.groups	local	
t rule.id	100200	
# rule.level	10	
🔔 rule.mail	false	
t rule.mitre.id	T1074	
t rule.mitre.tactic	Collection	
t rule.mitre.technique	Data Staged	
	t data.audit.file.name	/tmp/finaltest.txt
	t data.audit.fsgid	0
	t data.audit.fsuid	0
	t data.audit.gid	0
	t data.audit.id	436227
	t data.audit.key	tmp_watch
	t data.audit.pid	11458
	t data.audit.ppid	11457
	t data.audit.session	3
	t data.audit.sgid	0
	t data.audit.success	yes
	t data.audit.suid	0
	t data.audit.syscall	257
	t data.audit.tty	pts1
	t data.audit.type	SYSCALL
	t data.audit.uid	0
	t decoder.name	auditd
	t decoder.parent	auditd

Rule 2 — Brute Force Composite (10+ Failed Logins in 60 Seconds)

Objective: Detect 10 or more failed login attempts from the same source IP within 60 seconds.

Final rule:

```
xml
<rule id="100201" level="12" frequency="10" timeframe="60">
  <if_matched_sid>5760</if_matched_sid>
  <same_source_ip/>
  <description>SSH brute force: 10+ failed logins in 60 seconds from
same source</description>
  <mitre>
    <id>T1110.001</id>
  </mitre>
</rule>
```

```
fara@ubuntuerver:~$ cat > ~/day12-rule2.xml << 'EOF'
<group name="local">
  <rule id="100201" level="12" frequency="10" timeframe="60">
    <if_matched_sid>5760</if_matched_sid>
    <same_source_ip />
    <description>SSH brute force: 10+ failed logins in 60 seconds from same source</description>
    <mitre>
      <id>T1110.001</id>
    </mitre>
  </rule>
</group>
EOF

docker cp ~/day12-rule2.xml single-node-wazuh.manager-1:/var/ossec/etc/rules/day12_rule2.xml
docker exec single-node-wazuh.manager-1 /var/ossec/bin/wazuh-control start
Successfully copied 345B (transferred 2.05kB) to single-node-wazuh.manager-1:/var/ossec/etc/rules/day12_rule2.xml
2026/07/13 11:05:42 wazuh-modulesd:router: INFO: Loaded router module.
2026/07/13 11:05:42 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
Starting Wazuh v4.9.0 ...
wazuh-apid already running...
Started wazuh-csyslogd...
Started wazuh-dbd...
2026/07/13 11:05:44 wazuh-integrator: INFO: Remote integrations not configured. Clean exit.
Started wazuh-integrator...
Started wazuh-agentlessd...
wazuh-authd already running...
wazuh-db already running...
wazuh-execd already running...
wazuh-analysisd already running...
wazuh-syscheckd already running...
wazuh-remoted already running...
wazuh-logcollector already running...
wazuh-monitor already running...
wazuh-modulesd already running...
Completed.
```

Investigation summary: The rule was verified as structurally correct — its syntax was compared field-by-field against Wazuh's own official composite rule (5763), which uses an identical pattern (if_matched_sid, same_source_ip, frequency/timeframe) against the same base rule (5760). Over a dozen distinct hypotheses were tested with direct evidence:

- Threshold collision with rule 5763 (tested with frequency values both above and below 5763's threshold)
- Missing ignore attribute
- Incorrect rule chaining (if_sid vs if_matched_sid, tested against multiple parent rules)
- IP-correlation logic specifically (tested with same_source_ip removed entirely)
- Cross-validation against a completely independent data source (PAM/sudo authentication failures via rule 5401, reproducing the identical failure pattern)
- A wazuh-analysisd health degradation issue was discovered and fixed via full Manager container restart (confirmed via Wazuh API health check) — this resolved Rule 3's issue entirely but did not resolve Rule 2's
- An exact parameter match test against rule 5763's precise configuration (frequency=8, timeframe=120, ignore=60), including testing an alternate custom rule ID range (900100, matching previously successful deployments)

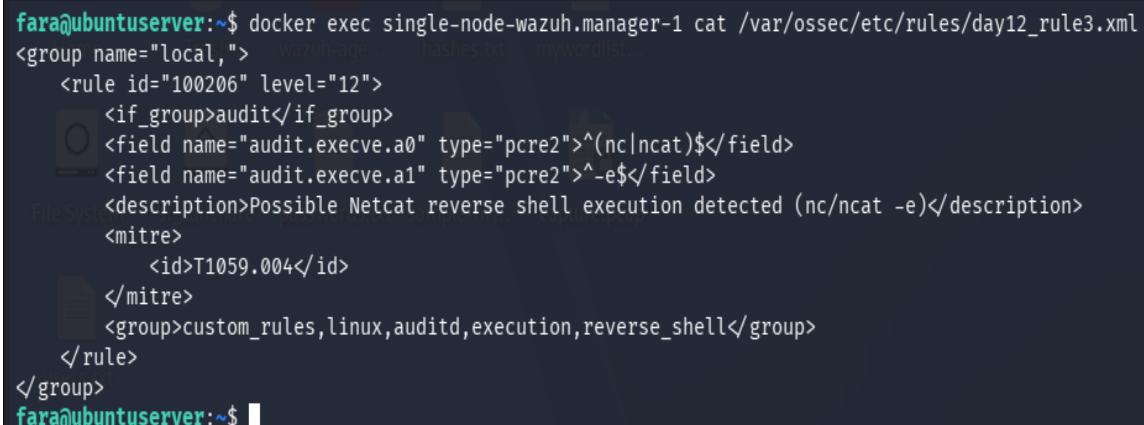
Conclusion: Despite the base rule (5760) firing correctly and Wazuh's own composite rule (5763) firing correctly on identical live data, the custom composite rule consistently did not fire under any tested configuration. The root cause remains undetermined after this investigation. Further validation on a clean Wazuh deployment and consultation with the Wazuh community are recommended to determine whether the behavior is environment-specific, configuration-related, or a product issue.

Rule 3 — Reverse Shell Indicator via auditd (nc/netcat -e)

Objective: Detect execution of netcat with the -e flag, a well-known reverse shell indicator.

Final rule:

```
<rule id="100206" level="12">
  <if_group>audit</if_group>
  <field name="audit.execve.a0" type="pcre2">^(nc|ncat)$</field>
  <field name="audit.execve.a1" type="pcre2">^-e$</field>
  <description>Possible Netcat reverse shell execution detected
(nc/netcat -e)</description>
  <mitre>
    <id>T1059.004</id>
  </mitre>
  <group>custom_rules,linux,auditd,execution,reverse_shell</group>
</rule>
```

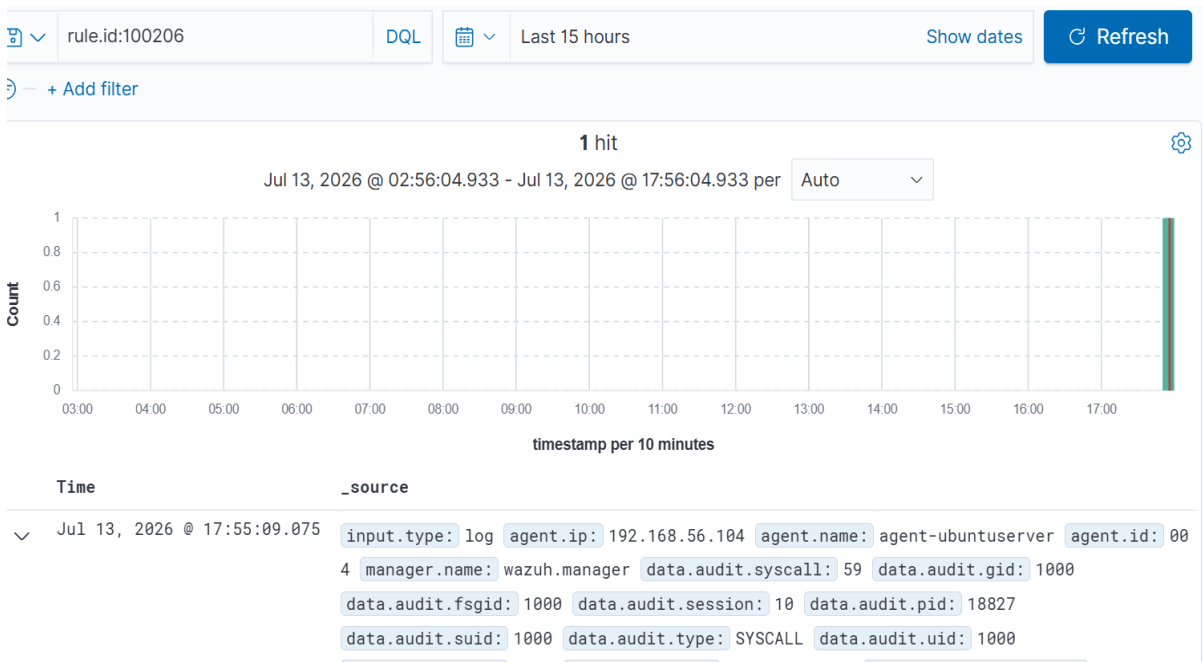


```
fara@ubuntuuserver:~$ docker exec single-node-wazuh.manager-1 cat /var/ossec/etc/rules/day12_rule3.xml
<group name="local,">
  <rule id="100206" level="12">
    <if_group>audit</if_group>
    <field name="audit.execve.a0" type="pcre2">^(nc|ncat)$</field>
    <field name="audit.execve.a1" type="pcre2">^-e$</field>
    <description>Possible Netcat reverse shell execution detected (nc/netcat -e)</description>
    <mitre>
      <id>T1059.004</id>
    </mitre>
    <group>custom_rules,linux,auditd,execution,reverse_shell</group>
  </rule>
</group>
fara@ubuntuuserver:~$
```

Key findings:

- Kali's default OpenBSD netcat does not support the -e flag; ncat (nmap package) was installed to generate a genuine test case.
- Real auditd EXECVE records were captured and used to precisely determine correct field names (audit.execve.a0, a1, a2) directly from Wazuh's official decoder source.
- Initial testing showed zero results despite correct rule syntax and confirmed source data; investigation traced this to the same wazuh-analysisd health degradation found while investigating Rule 2. A full Manager restart resolved this, and the rule fired correctly and cleanly on the very next live test.

Verification: Confirmed via live agent data — `ncat -e /bin/bash 127.0.0.1 4447` produced a single, clean, correctly-tagged alert matching all EXECVE arguments exactly.



t	full_log
	<pre> type=SYSCALL msg=audit(1783950898.876:643292): arch=c000003e syscall=59 success=yes exit=0 a0=55e98afeb590 a1=55e98afe98f0 a2=55e98b132a4 0 a3=8 items=2 ppid=12526 pid=18827 auid=1000 uid=1000 gid=1000 euid=1000 suid=1000 fsuid=1000 egid=1000 sgid=1000 fsgid=1000 tty=pts1 se s=10 comm="ncat" exe="/usr/bin/ncat" subj=unconfined key="process_creation"ARCH=x86_64 SYSCALL=execve AUID="fara" UID="fara" GID="fara" E UID="fara" SUID="fara" FSUID="fara" EGID="fara" SGID="fara" FSGID="fara" type=EXECVE msg=audit(1783950898.876:643292): argc=5 a0="ncat" a 1="-e" a2="/bin/bash" a3="127.0.0.1" a4="4447" type=CWD msg=audit(1783950898.876:643292): cwd="/home/fara" type=PATH msg=audit(178395089 8.876:643292): item=0 name="/usr/bin/ncat" inode=6189 dev=fd:00 mode=0100755 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 c ap_fe=0 cap_fm=0 cap_fd=0 cap_fm=0 rootid=0000 root=0 ogid=0 root" tunc=PATH msg=audit(1783950898.876:643292): item=1 name="/lib64/ld-linux-x86-64.s o" type=PATH msg=audit(1783950898.876:643292): item=2 name="/etc/passwd" type=PATH msg=audit(1783950898.876:643292): item=3 name="/etc/shadow" type=PATH msg=audit(1783950898.876:643292): item=4 name="/etc/group" type=PATH msg=audit(1783950898.876:643292): item=5 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=6 name="/etc/crontab" type=PATH msg=audit(1783950898.876:643292): item=7 name="/etc/passwd" type=PATH msg=audit(1783950898.876:643292): item=8 name="/etc/shadow" type=PATH msg=audit(1783950898.876:643292): item=9 name="/etc/group" type=PATH msg=audit(1783950898.876:643292): item=10 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=11 name="/etc/passwd" type=PATH msg=audit(1783950898.876:643292): item=12 name="/etc/shadow" type=PATH msg=audit(1783950898.876:643292): item=13 name="/etc/group" type=PATH msg=audit(1783950898.876:643292): item=14 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=15 name="/etc/passwd" type=PATH 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msg=audit(1783950898.876:643292): item=58 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=59 name="/etc/passwd" type=PATH msg=audit(1783950898.876:643292): item=60 name="/etc/shadow" type=PATH msg=audit(1783950898.876:643292): item=61 name="/etc/group" type=PATH msg=audit(1783950898.876:643292): item=62 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=63 name="/etc/passwd" type=PATH msg=audit(1783950898.876:643292): item=64 name="/etc/shadow" type=PATH msg=audit(1783950898.876:643292): item=65 name="/etc/group" type=PATH msg=audit(1783950898.876:643292): item=66 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=67 name="/etc/passwd" type=PATH msg=audit(1783950898.876:643292): item=68 name="/etc/shadow" type=PATH msg=audit(1783950898.876:643292): item=69 name="/etc/group" type=PATH msg=audit(1783950898.876:643292): item=70 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=71 name="/etc/passwd" type=PATH msg=audit(1783950898.876:643292): item=72 name="/etc/shadow" type=PATH msg=audit(1783950898.876:643292): item=73 name="/etc/group" type=PATH msg=audit(1783950898.876:643292): item=74 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=75 name="/etc/passwd" type=PATH msg=audit(1783950898.876:643292): item=76 name="/etc/shadow" type=PATH msg=audit(1783950898.876:643292): item=77 name="/etc/group" type=PATH msg=audit(1783950898.876:643292): item=78 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=79 name="/etc/passwd" type=PATH msg=audit(1783950898.876:643292): item=80 name="/etc/shadow" type=PATH msg=audit(1783950898.876:643292): item=81 name="/etc/group" type=PATH msg=audit(1783950898.876:643292): item=82 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=83 name="/etc/passwd" type=PATH msg=audit(1783950898.876:643292): item=84 name="/etc/shadow" type=PATH msg=audit(1783950898.876:643292): item=85 name="/etc/group" type=PATH msg=audit(1783950898.876:643292): item=86 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=87 name="/etc/passwd" type=PATH msg=audit(1783950898.876:643292): item=88 name="/etc/shadow" type=PATH msg=audit(1783950898.876:643292): item=89 name="/etc/group" type=PATH msg=audit(1783950898.876:643292): item=90 name="/etc/gshadow" type=PATH msg=audit(1783950898.876:643292): item=91 name="/etc/passwd" type=PATH msg=audit(1783950898.87</pre>

```
type=SYSCALL msg=audit(1783950898.876:643292): arch=c000003e syscall=59 success=yes exit=0 a0=55e98afeb590 a1=55e98afe9f0 a2=55e98b132a4  
a3=0 items=2 ppid=12526 pid=18827 uid=1000 gid=1000 uid=1000 gid=1000 syscall=1000 syscall=1000 fsgid=1000 tpts=se  
a3=10 comm="ncat" exe="/usr/bin/ncat" subj=unconfined key="process_creation" ARCH=x86_64 SYSCALL=execve AUID="fara" UID="fara" GID="fara" E  
UID="fara" SUID="fara" FSUID="fara" EGID="fara" SGID="fara" FSGID="fara" type=EXECVE msg=audit(1783950898.876:643292): argc=5 a0="ncat" a  
1="-e" a2="/bin/bash" a3="127.0.0.1" a4="4447" type=CWD msg=audit(1783950898.876:643292): cwd="/home/fara" type=PATH msg=audit(1783950898  
.876:643292): item=0 name="/usr/bin/ncat" inode=6189 dev=fd:0 mode=0100755 uid=0 ogid=0 rdev=0000 nametype=NORMAL cap_fp=0 cap_pi=0 c  
ap_fa=0 cap_fuer=0 cap_front=0BUID="root" OGID="root" type=PATH msg=audit(1783950898.876:643292): item=1 name="/lib64/libc.so.6" cap_fp=0 c
```

t id	1783950909.3935424
t input.type	log
t location	/var/log/audit/audit.log
t manager.name	wazuh.manager
t rule.description	Possible Netcat reverse shell execution detected (nc/ncat -e)
# rule.firedtimes	1
t rule.groups	local, custom_rules, linux, auditd, execution, reverse_shell
t rule.id	100206
# rule.level	12
rule.mail	true
t rule.mitre.id	T1059.004
t rule.mitre.tactic	Execution
t rule.mitre.technique	Unix Shell

†	data.audit.file.name	/usr/bin/ncat
†	data.audit.fsgid	1000
†	data.audit.fsuid	1000
†	data.audit.gid	1000
†	data.audit.id	643292
†	data.audit.key	process_creation
†	data.audit.pid	18827
†	data.audit.ppid	12526
†	data.audit.session	10
†	data.audit.sgid	1000
†	data.audit.success	yes
†	data.audit.suid	1000
†	data.audit.syscall	59
†	data.audit.tty	pts1
†	data.audit.type	SYSCALL
†	data.audit.uid	1000
†	decoder.name	auditd
†	decoder.parent	auditd

Deliverable

3 custom Wazuh XML rule files committed to Git repo:

- day12-rule1-privileged-tmp-file.xml
- day12-rule2-brute-force-composite.xml
- day12-rule3-reverse-shell-netcat.xml

Conclusion

Rule 1 surfaced two genuine bugs — a missing auditd watch rule and an overly permissive regex matching internal system noise — both identified and corrected through live-data verification rather than assumption. Rule 3's investigation led to the discovery of a wazuh-analysisd health degradation affecting the broader environment, which was diagnosed and resolved. Rule 2 remains a genuinely unresolved anomaly despite exceptionally thorough, evidence-based investigation spanning over a dozen tested hypotheses — a legitimate outcome that reflects real-world detection engineering, where not every issue resolves within a single session.